WORKED SOLUTION

STEP III 1999, Question 4

Pure • Elementary geometry

STEP III 1999, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q4

This question has a full guided treatment in *Beyond STEP*, Section 7.9, *Only Five Platonic Solids*: https://beyondstep.keithmcnulty.org/ch07/platonic-solids.html

The key idea

Count the edges in two different ways: once by going round the faces, and once by going round the vertices. Each count expresses E in terms of F (or V), and substituting into Euler's formula leaves a single equation for F. After that the whole classification comes down to one observation: F is positive, so h must be positive, and this leaves only a handful of whole-number pairs (m, n).

Solution

Deriving the formula for F

Each face is an m-gon, so it has m edges. Going round every face and counting its edges gives mF. But each edge is the boundary between exactly two faces, so this counts every edge twice:

$$mF = 2E.$$

In the same way, n edges meet at each vertex, so counting edges vertex by vertex gives nV. Each edge has exactly two endpoints, so this also counts every edge twice:

$$nV = 2E.$$

So $E = \frac{1}{2}mF$ and $V = \frac{2E}{n} = \frac{mF}{n}$. Substituting into $V - E + F = 2$ gives

$$\frac{mF}{n} - \frac{mF}{2} + F = 2$$

$$F\,(2m - mn + 2n) = 4n \qquad \text{(multiplying by } 2n\text{)}.$$

Now expand the given h:

$$4 - (n-2)(m-2) = 4 - (mn - 2m - 2n + 4) = 2m + 2n - mn,$$

which is exactly the bracket. So $Fh = 4n$. Here F and n are positive, so $h \neq 0$, and we can divide:

$$F = \frac{4n}{h}, \qquad h = 4 - (n-2)(m-2),$$

as required.

Only five regular polyhedra

Ask yourself What restrictions on m and n come from geometry rather than from the formula?

A polygon has at least three sides, so $m \geq 3$. At a vertex of a solid at least three faces must meet: two faces meeting along a corner would give a flat flap, not a corner of a solid. So $n \geq 3$ too. Then $m - 2 \geq 1$ and $n - 2 \geq 1$.

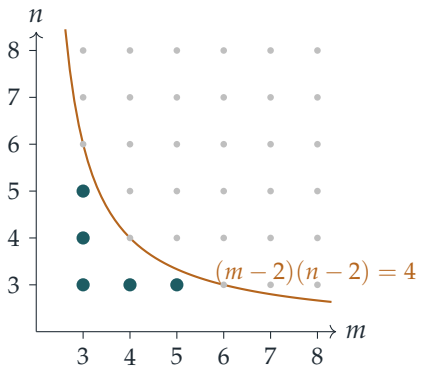
Since $F > 0$ and $4n > 0$, the formula forces $h > 0$, that is,

$$(m-2)(n-2) < 4.$$

The product of the two positive integers $m-2$ and $n-2$ is therefore 1, 2 or 3, so $h = 3$, 2 or 1. The only factorisations are

- $(m-2)(n-2) = 1$: $m = n = 3$, so $h = 3$;
- $(m-2)(n-2) = 2$: $(m,n) = (3,4)$ or $(4,3)$, so $h = 2$;
- $(m-2)(n-2) = 3$: $(m,n) = (3,5)$ or $(5,3)$, so $h = 1$.

The picture shows the same thing: the lattice points with $m, n \geq 3$ lying strictly inside the hyperbola $(m-2)(n-2) = 4$.



For each pair, $F = 4n/h$, $E = \frac{1}{2}mF$ and $V = mF/n$:

	m	n	h	V	E	F
tetrahedron	3	3	3	4	6	4
octahedron	3	4	2	6	12	8
cube	4	3	2	8	12	6
icosahedron	3	5	1	12	30	20
dodecahedron	5	3	1	20	30	12

For example, for the icosahedron $F = 4 \cdot 5/1 = 20$, $E = \frac{1}{2} \cdot 3 \cdot 20 = 30$ and $V = 3 \cdot 20/5 = 12$.

So at most five combinations of (m, n) are possible. Each one really does occur, as the five familiar Platonic solids named in the table show. So there are exactly five regular polyhedra.

Checks

- Every row satisfies $V - E + F = 2$. For the dodecahedron, $20 - 30 + 12 = 2$.
- The cube gives $V = 8$, $E = 12$, $F = 6$, which you can count on a die.
- Swapping m and n swaps V and F and keeps E fixed (cube and octahedron, dodecahedron and icosahedron). This is duality: put a vertex at the centre of each face.
- An independent check: at each vertex the angle deficit $2\pi - n\pi(m-2)/m$ must be positive, and it totals 4π over all V vertices (Descartes' theorem). This holds for all five, and a computer search over $3 \leq m, n < 200$ found no other pairs with $h > 0$.

Common slips

- Forgetting the factor of 2 when counting edges: each edge belongs to two faces and has two ends.
- Not saying why $n \geq 3$. Without it, $n = 2$ would give $h = 4$ for every m, and infinitely many "solutions".
- Concluding "exactly five" from the counting alone. The argument shows there are *at most* five; you need the actual solids to show that each case exists.

Beyond STEP

Euler's formula is the start of topology. The number $V - E + F$ (the *Euler characteristic*) is the same for every way of dividing a surface into faces, so it is 2 for a sphere, 0 for a torus, and so on. Repeating this argument on a torus gives infinitely many "regular" tilings, such as the square grid. The symmetry groups of the five solids are also central examples in a first group theory course. For a lively account, read Richeson, *Euler's Gem*; Armstrong, *Basic Topology*, takes the ideas further.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.